arg SOFTWARE. SOLUTIONS

Building Digital Solutions That Grow With You

We help innovative businesses move faster by building high-quality software with precision, confidence, and flexibility.

Who We Are

Technical Collaborators

We translate product ideas into real-world platforms that scale, ensuring each solution is purpose-built, easy to evolve, and grounded in business goals.

Adaptable Partners

We adapt to your needs - whether you need quick execution for a prototype or a more complex project. We think like product owners, not just engineers.

Clear Communicators

No fluff. No bloated meetings. Clear updates, written plans, and direct access to the engineers doing the work.



What We Do Best

We help companies move forward by building scalable software and shipping usable products. Our goal is always to provide reliable, elegant foundations that grow with your business.



Product Discovery & Prototyping

Aligning business goals with user needs to ensure the right product is built from the start.



MVP Development

Launch your product with just enough features to gather user feedback and validate your vision.



Frontend Development

Seamless interfaces across web and mobile, optimized for usability and performance.



Backend Development

Solid, well-structured APIs and business logic designed to handle real-world usage and scale.



Infrastructure

Cloud-native, containerized, and cost-efficient setups ready for growth, observability, and failover.

Our Founders



José Antunes

Website | LinkedIn

José Francisco Antunes is a senior software architect and co-founder of ARG Software with over a decade of experience designing and developing scalable, enterprise-grade applications. With a strong academic foundation from the University of Porto, José has worked across Europe and the U.S. in industries including music tech, manufacturing, ERP, tourism, fintech, and event-driven/real-time processing platforms.

He specializes in backend architecture, domain-driven design, and full-stack development using modern technologies such as .NET Core, Angular, React, PostgreSQL, and Docker. Known for his hands-on leadership style, José often acts as a Scrum Master and technical coach, fostering agile best practices across distributed teams.

A strong advocate for clean code and architecture patterns like CQRS and Onion Architecture, he is passionate about building reliable, maintainable systems that scale.



Rui Rocha

Website | LinkedIn

Rui Rocha is a seasoned full-stack software developer and co-founder of ARG Software, with over a decade of experience across diverse industries including telecom, music royalties, healthcare tech, food & brand design, fintech, and high-frequency/event-driven processing systems. His expertise spans front-end and back-end development using modern frameworks and tools such as React, Angular, .NET, Node.js, and Docker.

Rui is passionate about delivering scalable, robust solutions and excels in agile environments. With a strong foundation in software architecture, testing, and DevOps practices, he's led and contributed to projects in Portugal, Sweden, and the U.S.

Clean, modular design is a core priority, with an emphasis on building systems that are maintainable, testable, and scalable over time. Committed to delivering dependable and adaptable solutions, he brings a globally informed perspective to every team he joins.



Case Study Overview

We've delivered a variety of successful platforms across sectors. Here are a few standout examples selected for their architectural depth, product complexity, or potential to create lasting industry impact:

Mojaloop vNext

A real-time financial inclusion switch supporting regional payments across borders.

Dokutar

A user-friendly and regulation-aware tax documentation platform tailored for the German market.

SkyTracks

A browser-based music production platform with low-latency interactions and external instrument support.

More Case Studies

Royalty Flush

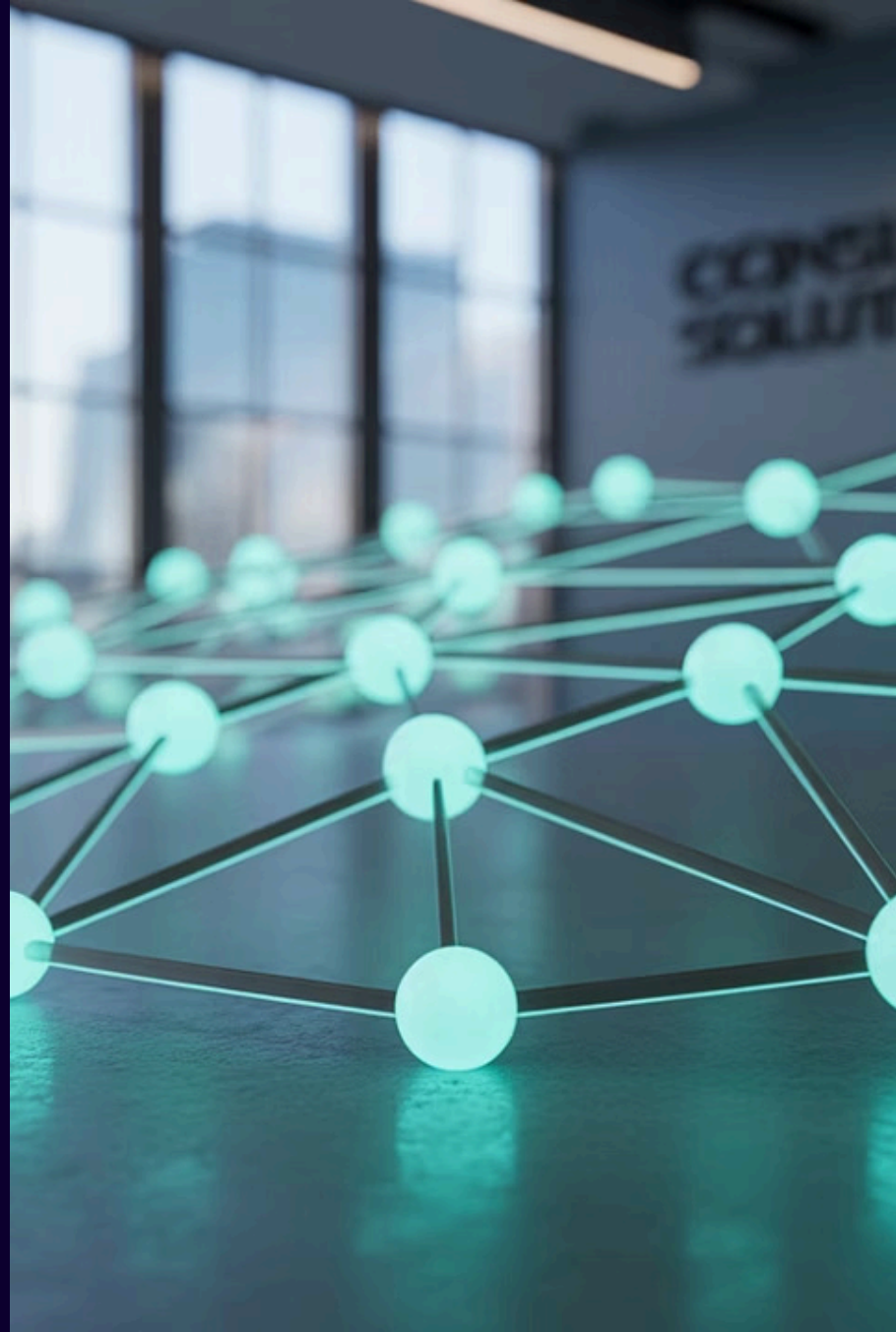
A royalty engine that processes CWR files, calculates obligations, and produces reports.

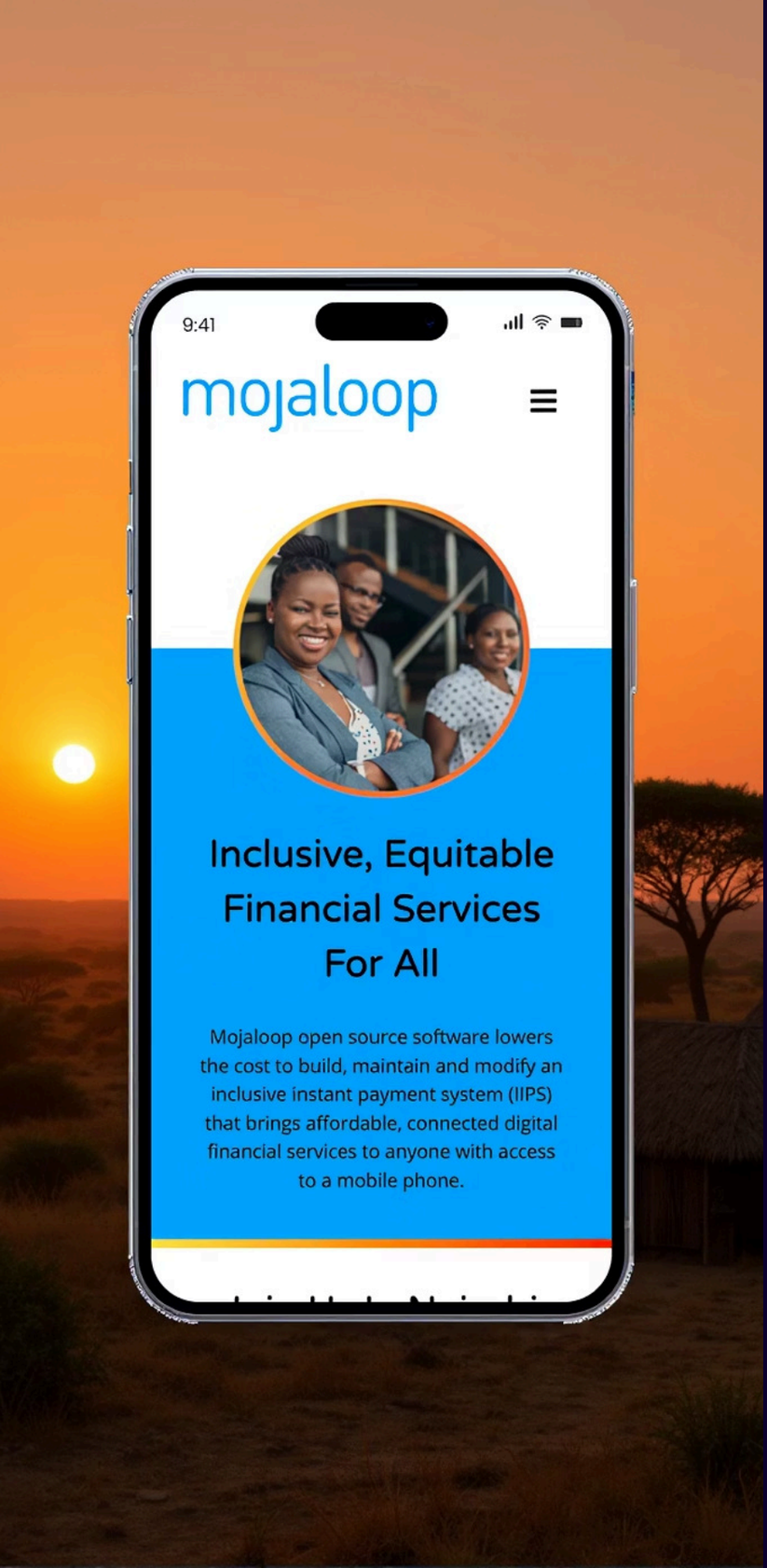
Vector

A trading platform allowing users to automate trades and subscribe to expert strategies.

People's Clearinghouse

An open-source digital clearing solution focused on community financial institutions.





Case Study: Mojaloop vNext

1

Challenge:

Mojaloop's mission to bring financial inclusion to underserved regions meant building a platform that could interconnect diverse financial systems across countries, while meeting high standards for performance, security, and regulation.

2

Our Solution:

- Joined the core team to redesign the switch as microservices
- Integrated real-time settlement capabilities for instant transactions
- Built robust authentication, encryption, and access control layers
- Created tools for handling regional regulatory compliance
- Supported modular onboarding of banks
- Implemented the central transaction switch using Domain-Driven Design

3

Impact:

Enables seamless digital payments across borders, empowering people in developing regions to access financial tools through mobile money, banking apps, and fintech solutions.

Case Study: Dokutar

Challenge:

Dokutar had outgrown its Node and WordPress foundation and needed strict data separation between clients.

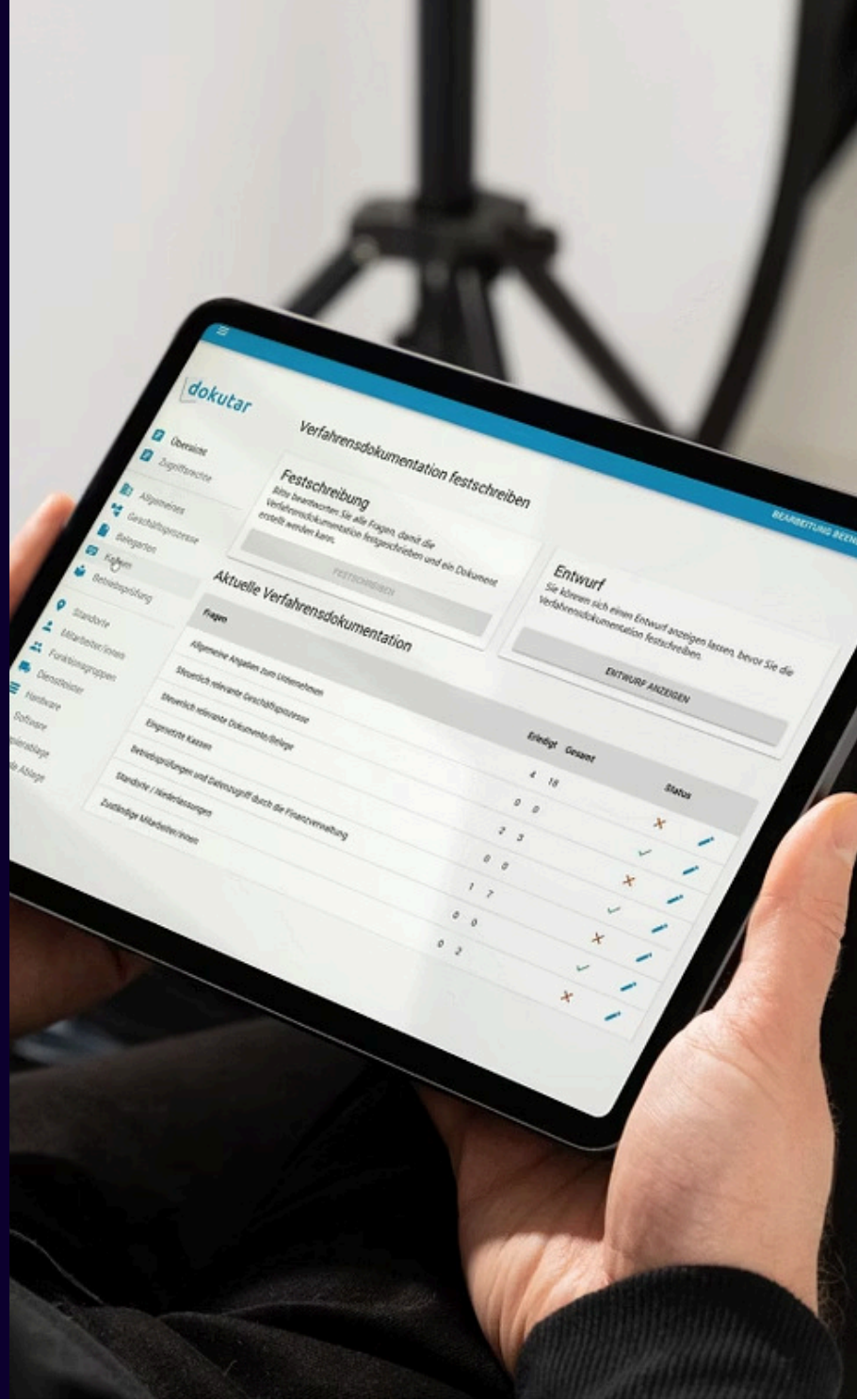
Impact:

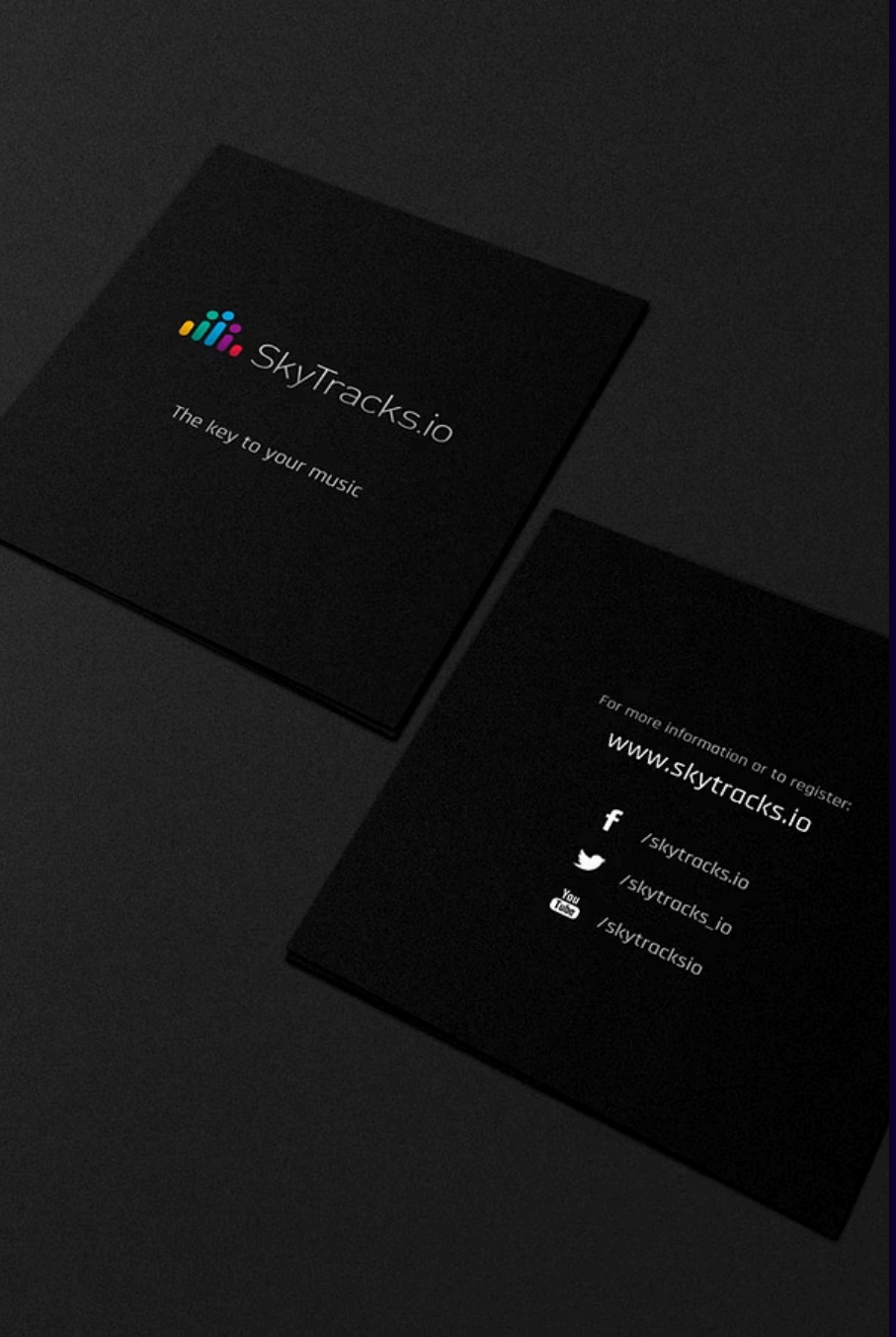
Dokutar gained a backend that supports more clients without compromising tenant privacy.

1 2 3

Our Solution:

- Redesigned the backend and database model so the platform could scale
- Planned and ran the migration from the legacy setup
- Introduced multitenancy and stronger security to keep each client's data separate





Case Study: SkyTracks

1

Challenge:

SkyTracks needed to launch a new version of its browser-based music production platform, with lower latency and reliable features.

2

Our Solution:

- Migrated the frontend to Angular with the new design
- Reduced data latency and connected external instruments
- Stabilized features that were unreliable before

3

Impact:

SkyTracks shipped its new version with an Angular frontend, lower-latency interactions, and external instrument support.

Case Study: Royalty Flush

1

Challenge:

Royalty processing ran on Excel sheets and macros that were hard to scale, audit, and trust as the data grew.

2

Our Solution:

- Built a backend royalty engine that ingests CWR files and calculates obligations
- Added reporting so royalty outputs are easy to review
- Moved repeatable logic out of manual spreadsheet macros

3

Impact:

Royalty operations moved from fragile spreadsheets to a repeatable workflow that is easier to run and review.





Case Study: Vector

Challenge:

Liminality Labs needed an automated trading platform that could connect user accounts on exchanges like Bybit and Coinbase and automate trades based on price signals and pre-defined strategies.

Our Solution:

- Rebuilt the entire backend from scratch with a new tech stack
- Integrated a reliable backend architecture to deliver full trading functionality and real-time strategy execution
- Enabled secure exchange integrations for automated crypto trades
- Restored full buy/sell automation based on user-defined price signals
- Implemented a robust strategies system allowing the users to subscribe to and run automated trading algorithms designed by experts

Impact:

Vector now provides a robust automation platform where users can subscribe to expert-built trading strategies that execute trades seamlessly.

Case Study: People's Clearinghouse

Challenge:

Expanding financial access across fragmented and underbanked communities required a new kind of digital clearinghouse - one that could process instant payments between grassroots financial providers using open protocols.

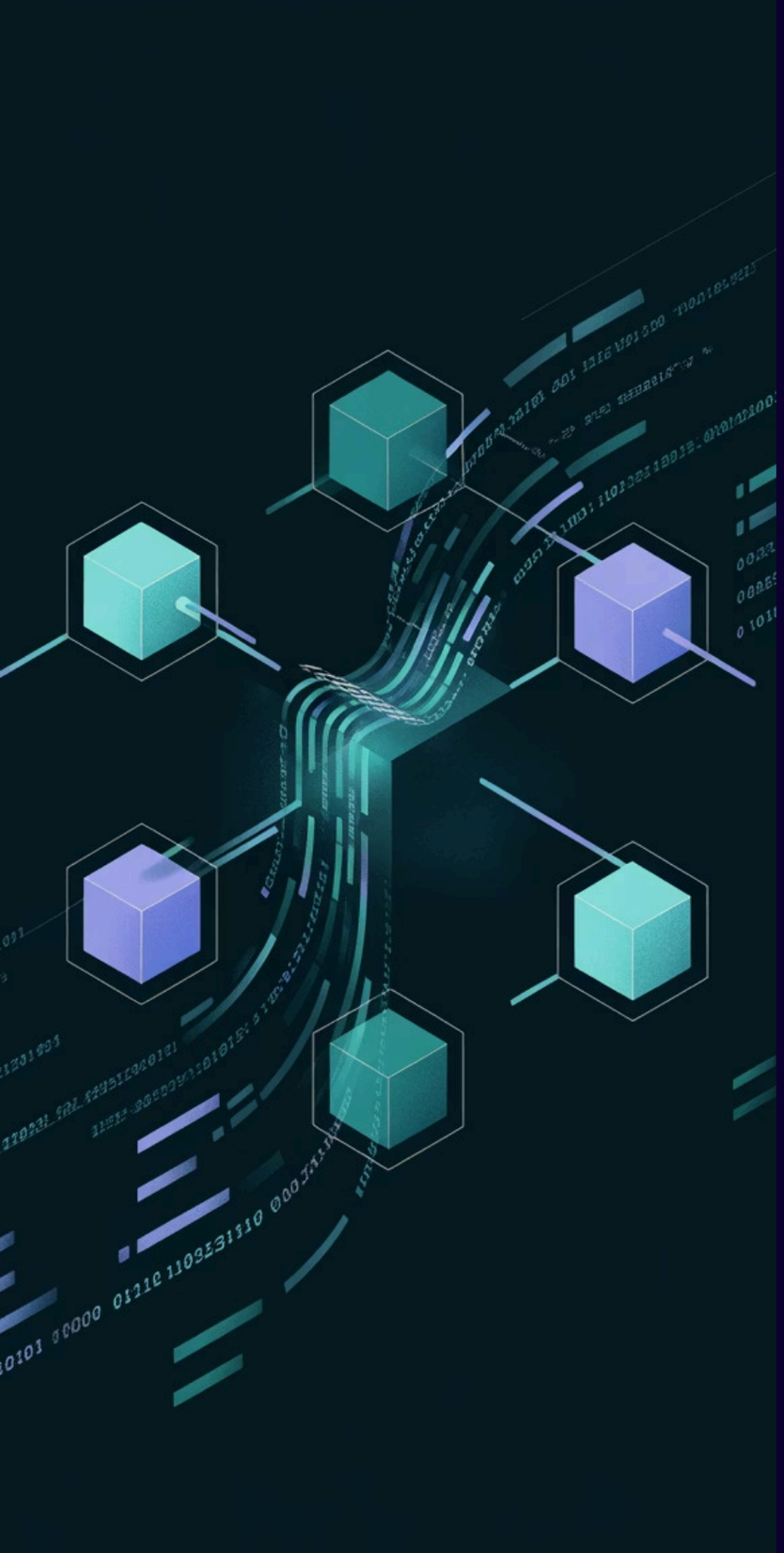
Impact:

This open-source solution serves as a foundation for national-scale clearing systems, offering a blueprint for inclusive infrastructure that can serve informal finance networks.

Our Solution:

- Built on Mojaloop vNext, with architecture led by Cyberbones
- Enabled interoperability between community banks and credit unions
- Implemented strong security and transparent accounting layers
- Focused on usability for rural, low-bandwidth environments





Our Open Source Projects

Nx-Monorepo-Boilerplate

A boilerplate for building scalable monorepo applications using Nx, featuring API integration, webhooks, messaging, and CQRS.

Browser Extension Boilerplate

Boilerplate for building Chrome extensions with React, TypeScript, and Vite - fast, modular, and ready to customize.

Clean-Architecture

A pragmatic Clean Architecture template for scalable, testable, and maintainable software.

Angular-Redux

An Angular project integrating Redux for state management, demonstrating a structured and modular approach to handling application state.

Kubernetes-Poc

A PoC for deploying a NestJS logging API with PostgreSQL in a Kubernetes cluster using K3d, showcasing deployments, services, and config management.

How We Work

Our Philosophy:

- We treat every project like it's our own
- We work in focused iterations
- We prioritize clarity and actionable communication
- We ship only working code

Process:

Discovery

Understand your goals and the problem space.

Planning

Create a roadmap with priorities and outcomes.

Development

Build with constant validation and iteration.

Testing

Validate reliability and behavior before shipping.

Launch & Support

Help you scale, monitor, and evolve the product.





Why Clients Love Working With Us

> "It was a pleasure to work with! Good communication, solid work..."
> Marc-Andre Mignault, Project Manager at SkyTracks

> "ARG Software were great to work with from the first day."
> Andre Mendo, CEO at Hostelier

> "They transformed a legacy API from PHP to TypeScript..."
> Hendrik, CTO at Dokutar

> "ARG did great! Was awesome to work with and always quick to respond."
> Austin Klise, CEO at Klise Consulting

We've successfully delivered critical systems that power production environments, handle significant transaction volumes, and serve diverse clients around the world.

Let's Talk

Have an idea? Whether it's an MVP or a complex platform, we're here to help



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Connect With Us

Website

Explore our official website for more information about our services and portfolio

LinkedIn

Connect with our team and stay updated on our latest projects

GitHub

Check out our open-source projects and contributions

Medium

Follow our technical insights and industry perspectives